

DIP QUARTZ CRYSTAL RESONATOR

石英晶体谐振器



Dip Crystal 11.4×4.8 mm HC-49U/S

- Package Height 3.5 mm max, 2.5~3.5 mm available
- Wave Soldering Temperature: 265°C max
- highly stable and reliable crystal unit with a metallic package
- Taped Version Available

RoHS
2002/95/EC

RoHS compliant

★ PARAMETERS

技术参数

PARAMETERS	参数	SPECIFICATION 规格
Frequency Range	频率范围	3.5~100MHz
Operation Mode	振动模式	(3.5 ~ 40MHz) Fundamental (27 ~ 100MHz) 3rd Overtone
Loading Capacitance	负载电容	20pF Std. 8 to 33pF, Series available
Drive Level	激励电平	100 μ W (500 μ W max)
Frequency Tolerance	频率偏差	± 10 ppm ~ ± 30 ppm (at 25°C)
Equivalent Resistance	谐振电阻	See Below
Frequency Stability	频率稳定性	± 2.5 ppm ~ ± 50 ppm
Operating Temp. Range:	工作温度范围	0 ~ +50°C to -40 ~ +85°C
Storage Temp. Range:	储存温度范围	-55 ~ +125°C

○ All specification subject change without notice. 规格变化，恕不另行通知。

★ FREQUENCY STABILITY VS. TEMPERATURE

频率温度特性

Operation Temperature Range	Frequency Stability					
	± 2.5 ppm	± 5 ppm	± 10 ppm	± 20 ppm	± 30 ppm	± 50 ppm
0°C ~ +50°C	○	○	○	●	○	○
-10°C ~ +60°C		○	○	●	○	○
-20°C ~ +70°C			○	○	●	○
-40°C ~ +85°C				○	○	●

● standard ○ available

★ ESR (SERIES RESISTANCE RS)

Frequency	Vibration Mode	ESR
3.2-3.999MHz	AT CUT/FUND.	150 Ω (max)
4.000-4.499MHz	AT CUT/FUND.	120 Ω (max)
4.500-5.999MHz	AT CUT/FUND.	100 Ω (max)
6.000-7.999MHz	AT CUT/FUND.	80 Ω (max)
8.000-9.999MHz	AT CUT/FUND.	60 Ω (max)
10.000-11.999MHz	AT CUT/FUND.	50 Ω (max)
12.000-13.999MHz	AT CUT/FUND.	40 Ω (max)
14.000-40.000MHz	AT CUT/FUND.	30 Ω (max)
27.000-100MHz	AT 3rd /OT	100 Ω (max)

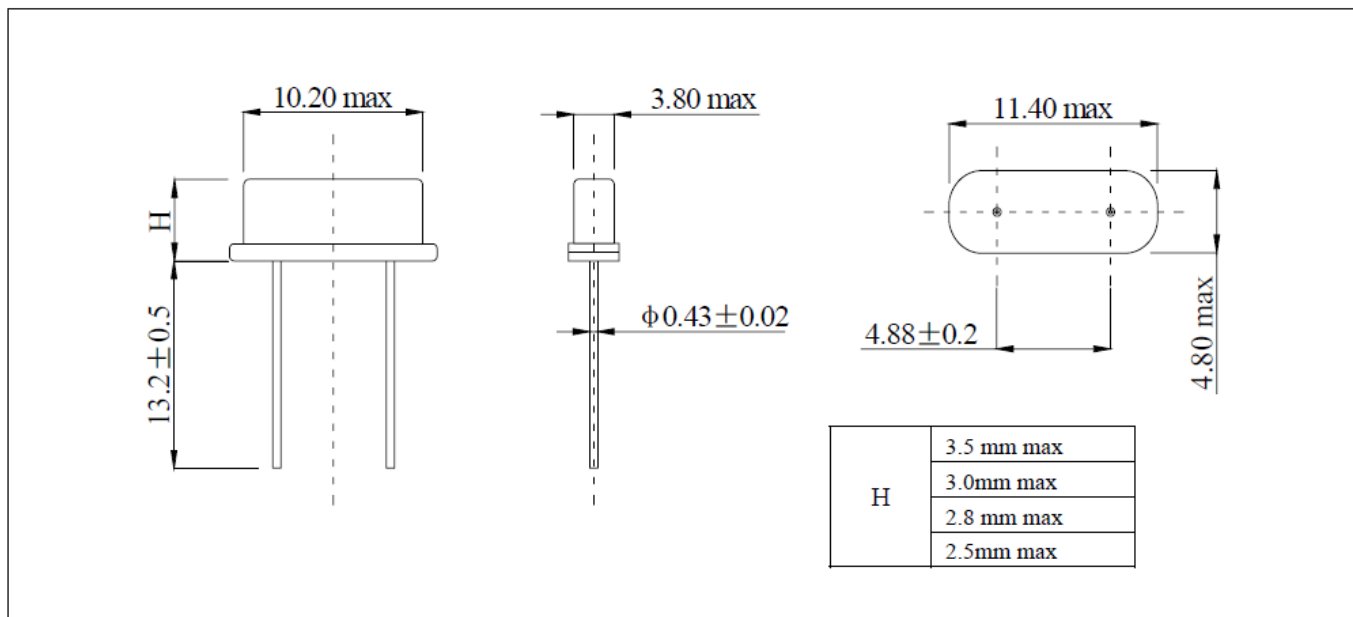
★ PART NUMBER GUIDE 部件号示例

e.g. FTX16.000M20S-30/30B (*S=HC49/S TYPE)

Logo	Quartz Crystal Resonator 石英晶体谐振器	Frequency 频率 Hz	Load Capacitance 负载电容 pF	Package 盒型	Frequency Tolerance 常温频差 ppm	Frequency Stability 温度频差 ppm	Operating Temp. Range 工作温度
FT	X	16.000M	20	S	30	30	B

Definition	Description
Operating Temperature Range	A: -10~+60°C
	B: -20~+70°C
	C: -30~+80°C
	D: -40~+85°C
	E: Customer specified

★ DIMENSIONS (Unit: mm) 外形尺寸



★ WAVE SOLDERING PROFILE 波峰焊特性

